MILITARY SPECIFICATION

SWITCHING GROUP, TRANSMITTER ANTENNA OK-261/BRC

This amendment forms a part of Military Specification MIL-S-28854(EC), dated 19 February 1980, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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* 2.1, "SPECIFICATIONS", "MILITARY": Delete "MIL-E-15090 Enamel, Equipment, Light-Gray (Formula No. 111)" and substitute "MIL-C-5015 Connector, Electrical, Circular, Threaded, AN Type, General Specification For".

* 2.1, "SPECIFICATIONS", "MILITARY", after "MIL-C-24308", Add:

"MIL-C-26482	Connector, Electrical (Circular, Miniature Quick Disconnect Environment Resistant) Receptacles And Plugs, General Specification For"

* 2.1, "SPECIFICATIONS", "MILITARY": Delete "MIL-C-28777/1" added by Amendment 1.

* 2.1, "SPECIFICATIONS", "MILITARY", after "MIL-C-26482", Add:

"MIL-C-28840	Connectors, Electrical, Circular, Threaded High Density, High Shock, Shipboard Class D, General Specification For".

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* 2.1, "STANDARDS", "MILITARY": Delete "MIL-STD-740 Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment".

* 2.1, "STANDARDS", "MILITARY": Delete "MIL-STD-965 Parts Control Program".

* 2.1, "STANDARDS", "MILITARY", after "MIL-STD-1472", Add:

"MS 3057	"Clamp, Cable, Electrical Connector"

* 3.1: Delete the text and substitute: "The equipment shall be in accordance with MIL-E-16400 to the extent specified herein."

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* 3.2.1: Add after paragraph title: "Performance characteristics and RF characteristics shall be verified at the RF interface connector panel on the rear of the equipment."

* 3.2.1.4, line 2: Delete "0.1" and "0.2" and substitute "0.16" and "0.44", respectively.

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* 3.2.1.9, line 1: Delete "1.1:1" and "1.2:1" and substitute "1.15:1" and "1.3:1", respectively.

* 3.2.1.11, line 3: Delete "a subpanel visible through the front panel." and substitute "the front panel and on a subpanel.".

* 3.2.1.11, line 4: Delete "this" and substitute "the".

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* 3.2.1.13, last sentence: Delete and substitute: "The interlock circuits shall open at least 20 milliseconds (ms) prior to RF switch opening, and shall close after the RF switch closure.".

* 3.2.1.14, line 2: Add "illuminated" before "green".

* 3.2.1.14, last sentence: Delete.

* 3.2.1.15, line 1: Delete "Each row of the 5 by 6" and substitute "Each horizontal row of the".

* 3.2.1.15, line 3: Delete "be in" and substitute "operate to".

* 3.2.1.16.1: Delete and substitute:

"3.2.1.16.1 Power cable connector. The input power connector on the equipment shall be a Type MS3102R14S-7P. The connector shall mate with a Type MS3106A14S-7S connector with a MS3057-6A clamp to be supplied with the equipment. The alternating current (AC) power connector shall have the safety ground lead connected to pin B, and the AC leads connected to pins A and C.".

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* FIGURE 2: Delete and substitute the attached new FIGURE 2.

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* 3.2.2.2: Delete the text and substitute: "A protective cover shall be provided to protect the front panel manual override controls of the RF switch matrix. The design of the cover shall provide easy access and removal. Gaskets shall not be required for the protective cover.".

* 3.2.2.3, line 3: After "and" add "a minimum of 2.5 cm (1 in.)".

* 3.2.3.1, third sentence: Delete and substitute: "Individual RF switching components shall have a life expectancy of at least 50,000 crosspoint operations without maintenance or overhaul. A crosspoint operation shall be defined as a 180 degree crosspoint rotation consisting of the 90 degree rotation from the THRU to TURN position to establish an RF connection plus the 90 degree rotation from the TURN to THRU position to clear the RF connection.".

* 3.2.4.1, line 2: Delete "0.5" and substitute "1.0"; line 3: Delete "1.5" and substitute "2.0".

* 3.2.4.1, last line: Delete "allows degraded operation." and substitute "results in degraded performance.".

* 3.2.4.2: Delete.

* 3.2.5.2: Delete.

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* 3.3.1: Delete.

* 3.3.1.1, first sentence: Delete and substitute: "All RF connectors utilized in the switching matrix shall be functionally equivalent to Type N as specified in MIL-C-39012.".

* 3.3.1.1, second sentence: Delete and substitute: "Mating connectors for the control, interlock, and AC power connectors shall be furnished. Control connectors shall be in accordance with MIL-C-28840, interlock connectors shall be in accordance with MIL-C-26482, and AC power connectors shall be in accordance with MIL-C-5015. Mating connectors for the RF cables will not be furnished.".

* 3.3.1.2: Delete the text and substitute: "The exterior finish of the equipment shall conform to the requirements of MIL-E-16400. An alternate exterior finish method shall be used only upon approval of the procuring activity.".

* 3.3.1.3.2, line 4: Delete "62.5 cm (25 ft)" and substitute "13.72-meter (45-foot)".

* 3.3.1.3.2, line 6: after last sentence: Add: "The cable shall be supplied with one connector installed and the other connector supplied loose.".

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* 3.3.2: Delete the text and substitute: "The switch matrix and remote control unit shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class A5 equipment with the exception that RS03 shall be performed at 10 volts per meter. If line-to-ground powerline filters are used to conform to the MIL-STD-461 requirements, the total capacitance shall not exceed 0.1 microfarad (μF) for 60 Hz equipments.

* 3.3.2.1: Delete.

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* FIGURE 3: Delete.

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* FIGURE 4: Delete

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* 3.3.2.2: Delete and substitute:

"3.3.2.2 Transients caused by keying transmitters and power supply induced noise. Transients induced in one circuit due to the keying of a transmitter in another circuit, shall not exceed 7 decibels referred to 1 watt (dBW) peak. In addition, power supply related noise induced into any circuit shall not exceed 7 decibels referred to 1 milliwatt (dBm) peak.".

After "3.3.6", add the following new paragraph:

"3.3.7 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400.".

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* TABLE II: Delete "Parts control" entry in its entirety.

* TABLE II: After "Characteristic impedance": "VSWR": Add "X" in "Group A" column.

* TABLE II: After "Isolation": Delete "VSWR" entry in its entirety.

* TABLE II: Delete "Airborne and structureborne noise" entry in its entirety.

* TABLE II: "Enclosure": Add "X" in "First article inspection" column.

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4.5.2.2, second sentence: Delete and substitute "The number of cycles required shall be 10 cycles." and in line 7, after "failure-free." insert the following new sentence: "When performance measurements are required, a functional operating test shall be performed.".

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TABLE III: Delete.

* After 4.5.5, add the following new paragraphs:

"4.5.5.1 Induced transients due to keying test. With one circuit passively loaded 50 ohms in and 50 ohms out, a transmitter at full power in an adjacent circuit shall be manually keyed on and off. The passive circuit, monitored with an oscilloscope, shall not display a response in excess of the limit specified in 3.3.2.2. This shall be performed with a transmitter in each of the frequency bands intended for use.

"4.5.5.2 Power supply induced noise. The passively loaded circuit of 4.5.5.1 shall not display power supply related noise (harmonics and so forth) in excess of the limit specified in 3.3.2.2.".

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* 4.6: Delete the text and substitute: "Reliability testing shall be performed in accordance with MIL-STD-781 Test Plan IVC. Test temperature shall be maintained at 50°C. Vibration stress shall be as specified in the Combined environments for shipboard equipment paragraph of MIL-STD-781. The period of vibration, equipment duty cycle, and AC line voltage shall be as shown in FIGURE 8. The operational duty cycle shall be 20 crosspoint switching operations per 24-hour period. Each crosspoint shall be sequentially operated from the THRU to TURN position and then from the TURN to THRU position and the proper operation of each crosspoint shall be verified for each position by monitoring the corresponding indicator and interlock circuits.".

* 4.7.1, line 3: After "M_{ct}" add "and M_{maxct}".

* 4.7.3: Delete.

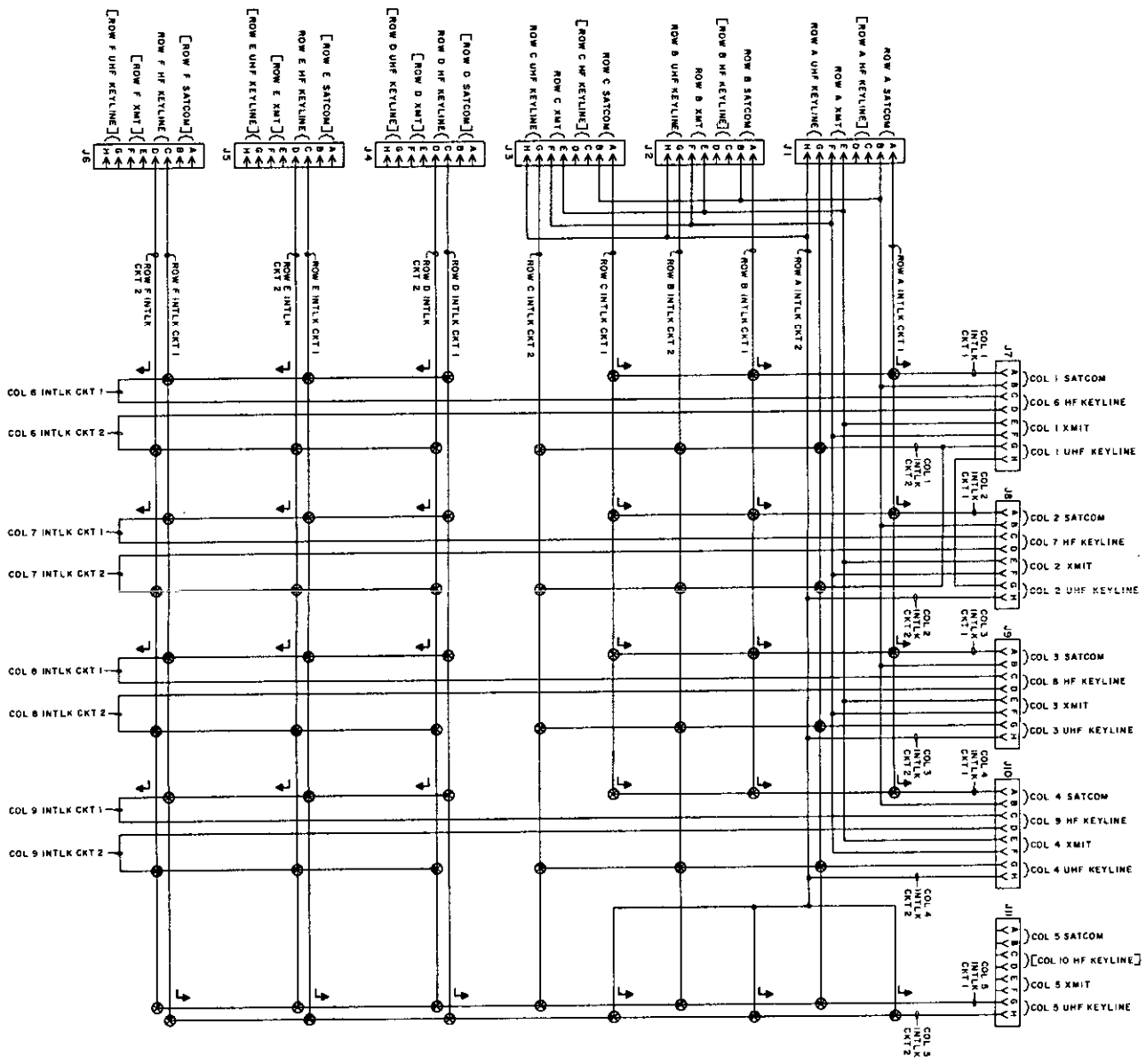
* Add the attached new FIGURE 8.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
NAVY-EC

(Project 5820-N777)

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NOTES:
1. HF KEYLINE CONNECTS TO PINS G AND H OF J11 SINCE COLUMN 5 IS NOT SPLIT
2. PIN J, K, AND L IN CONNECTORS J1 THROUGH J11 ARE SPARE AND ARE NOT SHOWN
3. ⊗ INDICATES INTERLOCK CIRCUIT SWITCHING

FIGURE 2. Switch matrix interlock control switching.

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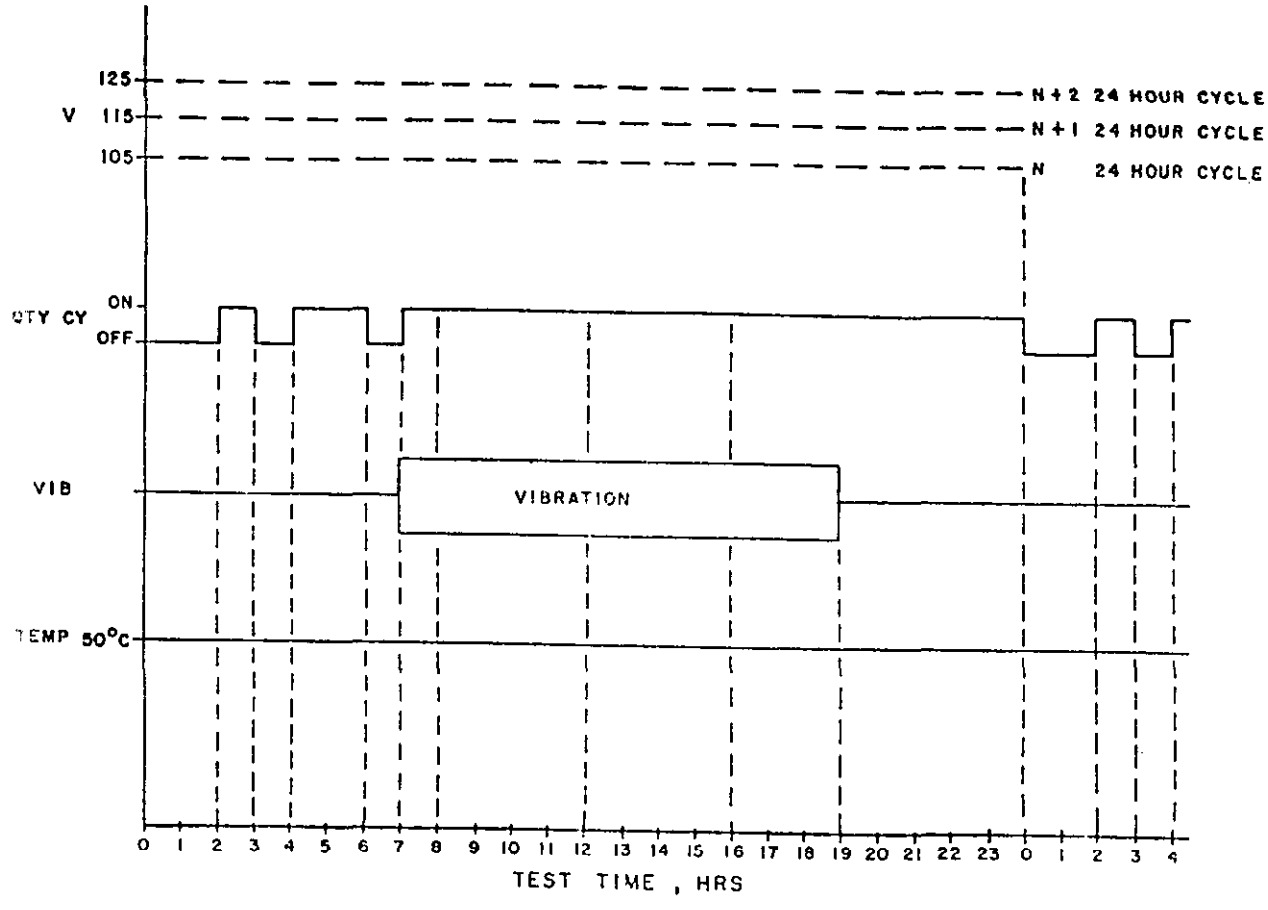


FIGURE 8. Test parameters for conducting reliability demonstration.

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